Import Dataset

In [1]:
```
import pandas as pd
```

In [2]:
```
car = pd.read_csv('D:/Downloads/Udemy-PY/Project+2+-+Cars+Dataset.csv')
```

In [3]:
```
car
```

Out[3]:

	Make	Model	Type	Origin	DriveTrain	MSRP	Invoice	EngineSize	Cylinders
0	Acura	MDX	SUV	Asia	All	$36,945	$33,337	3.5	6.0
1	Acura	RSX Type S 2dr	Sedan	Asia	Front	$23,820	$21,761	2.0	4.0
2	Acura	TSX 4dr	Sedan	Asia	Front	$26,990	$24,647	2.4	4.0
3	Acura	TL 4dr	Sedan	Asia	Front	$33,195	$30,299	3.2	6.0
4	Acura	3.5 RL 4dr	Sedan	Asia	Front	$43,755	$39,014	3.5	6.0
...	...	...	...	...	...	...	...	...	...
427	Volvo	C70 LPT convertible 2dr	Sedan	Europe	Front	$40,565	$38,203	2.4	5.0
428	Volvo	C70 HPT convertible 2dr	Sedan	Europe	Front	$42,565	$40,083	2.3	5.0
429	Volvo	S80 T6 4dr	Sedan	Europe	Front	$45,210	$42,573	2.9	6.0
430	Volvo	V40	Wagon	Europe	Front	$26,135	$24,641	1.9	4.0
431	Volvo	XC70	Wagon	Europe	All	$35,145	$33,112	2.5	5.0

432 rows × 15 columns

Checking Metadata of th Dataset

In [4]:
```
car.shape
```

Out[4]: (432, 15)

In [13]:
```
car.head(2)
```

Out[13]:

	Make	Model	Type	Origin	DriveTrain	MSRP	Invoice	EngineSize	Cylinders	Horse
0	Acura	MDX	SUV	Asia	All	$36,945	$33,337	3.5	6.0	
1	Acura	RSX Type S 2dr	Sedan	Asia	Front	$23,820	$21,761	2.0	4.0	

In [6]:

```
car.isnull().sum()
```

Out[6]:

```
Make           4
Model          4
Type           4
Origin         4
DriveTrain     4
MSRP           4
Invoice        4
EngineSize     4
Cylinders      6
Horsepower     4
MPG_City       4
MPG_Highway    4
Weight         4
Wheelbase      4
Length         4
dtype: int64
```

In [11]:

```
car_cleaned = car.dropna( axis = 0 )
#car['Cylinders'].filna(car['Cylinders'].mean() inplace = True) --to fill the null
```

In [12]:

```
car_cleaned.isnull().sum()
```

Out[12]:

```
Make           0
Model          0
Type           0
Origin         0
DriveTrain     0
MSRP           0
Invoice        0
EngineSize     0
Cylinders      0
Horsepower     0
MPG_City       0
MPG_Highway    0
Weight         0
Wheelbase      0
Length         0
dtype: int64
```

1. Find the total counts of make by category

In [15]:

```
car_cleaned['Make'].value_counts()
```

Out[15]:
```
Make
Toyota           28
Chevrolet        27
Mercedes-Benz    26
Ford             23
BMW              20
Audi             19
Nissan           17
Honda            17
Chrysler         15
Volkswagen       15
Mitsubishi       13
Dodge            13
Volvo            12
Hyundai          12
Jaguar           12
Pontiac          11
Kia              11
Subaru           11
Lexus            11
Mazda             9
Buick             9
Mercury           9
Lincoln           9
Cadillac          8
GMC               8
Saturn            8
Suzuki            8
Infiniti          8
Acura             7
Saab              7
Porsche           7
Land Rover        3
Oldsmobile        3
Jeep              3
Isuzu             2
MINI              2
Scion             2
Hummer            1
Name: count, dtype: int64
```

2. List only origins of Asia and Europe

In [17]:
```
car_cleaned[car_cleaned['Origin'].isin(['Asia','Europe'])]
```

Out[17]:

	Make	Model	Type	Origin	DriveTrain	MSRP	Invoice	EngineSize	Cylinders
0	Acura	MDX	SUV	Asia	All	$36,945	$33,337	3.5	6.0
1	Acura	RSX Type S 2dr	Sedan	Asia	Front	$23,820	$21,761	2.0	4.0
2	Acura	TSX 4dr	Sedan	Asia	Front	$26,990	$24,647	2.4	4.0
3	Acura	TL 4dr	Sedan	Asia	Front	$33,195	$30,299	3.2	6.0
4	Acura	3.5 RL 4dr	Sedan	Asia	Front	$43,755	$39,014	3.5	6.0
...	...	...	...	...	...	...	...	...	...
427	Volvo	C70 LPT convertible 2dr	Sedan	Europe	Front	$40,565	$38,203	2.4	5.0
428	Volvo	C70 HPT convertible 2dr	Sedan	Europe	Front	$42,565	$40,083	2.3	5.0
429	Volvo	S80 T6 4dr	Sedan	Europe	Front	$45,210	$42,573	2.9	6.0
430	Volvo	V40	Wagon	Europe	Front	$26,135	$24,641	1.9	4.0
431	Volvo	XC70	Wagon	Europe	All	$35,145	$33,112	2.5	5.0

279 rows × 15 columns

3. Remove all rows where weight is greater than 4000

In [20]:
```
car_cleaned[~(car_cleaned['Weight'] > 4000)]
```

Out[20]:

	Make	Model	Type	Origin	DriveTrain	MSRP	Invoice	EngineSize	Cylinde
1	Acura	RSX Type S 2dr	Sedan	Asia	Front	$23,820	$21,761	2.0	
2	Acura	TSX 4dr	Sedan	Asia	Front	$26,990	$24,647	2.4	
3	Acura	TL 4dr	Sedan	Asia	Front	$33,195	$30,299	3.2	
4	Acura	3.5 RL 4dr	Sedan	Asia	Front	$43,755	$39,014	3.5	
5	Acura	3.5 RL w/Navigation 4dr	Sedan	Asia	Front	$46,100	$41,100	3.5	
...	...	...	...	...	...	...	...	...	
427	Volvo	C70 LPT convertible 2dr	Sedan	Europe	Front	$40,565	$38,203	2.4	
428	Volvo	C70 HPT convertible 2dr	Sedan	Europe	Front	$42,565	$40,083	2.3	
429	Volvo	S80 T6 4dr	Sedan	Europe	Front	$45,210	$42,573	2.9	
430	Volvo	V40	Wagon	Europe	Front	$26,135	$24,641	1.9	
431	Volvo	XC70	Wagon	Europe	All	$35,145	$33,112	2.5	

323 rows × 15 columns

In [24]:
```
car_cleaned.loc[car_cleaned['MPG_City'] > 25, 'MPG_City'] = car_cleaned['MPG_City']
```

In [25]:
```
car_cleaned.head(10)
```

Out[25]:

	Make	Model	Type	Origin	DriveTrain	MSRP	Invoice	EngineSize	Cylinders
0	Acura	MDX	SUV	Asia	All	$36,945	$33,337	3.5	6.0
1	Acura	RSX Type S 2dr	Sedan	Asia	Front	$23,820	$21,761	2.0	4.0
2	Acura	TSX 4dr	Sedan	Asia	Front	$26,990	$24,647	2.4	4.0
3	Acura	TL 4dr	Sedan	Asia	Front	$33,195	$30,299	3.2	6.0
4	Acura	3.5 RL 4dr	Sedan	Asia	Front	$43,755	$39,014	3.5	6.0
5	Acura	3.5 RL w/Navigation 4dr	Sedan	Asia	Front	$46,100	$41,100	3.5	6.0
6	Acura	NSX coupe 2dr manual S	Sports	Asia	Rear	$89,765	$79,978	3.2	6.0
7	Audi	A4 1.8T 4dr	Sedan	Europe	Front	$25,940	$23,508	1.8	4.0
8	Audi	A41.8T convertible 2dr	Sedan	Europe	Front	$35,940	$32,506	1.8	4.0
9	Audi	A4 3.0 4dr	Sedan	Europe	Front	$31,840	$28,846	3.0	6.0

In [ ]: